Sihe (Julie) Wang

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EDUCATION

University of Virginia, Charlottesville, VA **May 2027**

BA in Cognitive Science with a Computer Science concentration, Minor in Data Science, and Drama **GPA 3.85**

SKILLS

Technical Skills & AI/ML: Python (NumPy, Pandas, Scikit-learn), Java, SQL, R, MATLAB, HTML5/CSS

Research Methods: Quantitative & Qualitative Research, Experimental Design, User Modeling, Heuristic Evaluation, A/B Testing

Design & Prototyping Tools: Figma (Advanced Prototyping), Adobe Creative Suite, Qualtrics, Maze, TouchDesigner

RESEARCH EXPERIENCE

UVA School of Data Science, Human-AI Design Lab, *Research Assistant* **June 2026 – present**

- Conducting a field study of **radiology reporting workflows** at UVA Health under the supervision of Prof. Nur Yildirim, examining how radiologists use hyperlinked reporting (Philips ViewPacks) through in-situ observation and semi-structured interviews

Harvard FAS CAMlab, *Digital Media & Research Intern*, Boston, MA **July - August 2025**

- Conducted **epigraphic and spatial analysis** of Xiangtangshan Cave (Hebei, China) Buddhism inscriptions to support scholarly interpretation, immersive reconstruction, and ongoing exhibition preparation
- Collaborated with art historians and media artists to develop historically grounded **XR experiences** using **TouchDesigner** and visual archive materials

UVA Neurosurgery Department, Human Brain Function Lab, *Research Assistant* **August 2024 - June 2025**

- Developed and optimized automated physiological signal processing pipelines in **MATLAB and Python** to analyze EEG, ECG (EKG), and laser-evoked potentials across 200+ clinical sessions under the supervision of **Prof. Jeff Liu**
- Co-authored findings demonstrating the posterior insula's role in heart rate modulation; results were disseminated through **two poster presentations**, and participated in the UVA Brain Retreat Conference

Tsinghua Laboratory of Brain and Intelligence, *Research Assistant,* Beijing, China **June - July 2024**

- Assisted in developing and executing an EEG and fNIRS field study investigating neural responses to live traditional Chinese music performances under the supervision of Prof. Xiaoqin Wang and Prof. Juan Huang
- Facilitated data collection and signal preprocessing for diverse participant groups, ensuring high data quality under naturalistic listening environments

DESIGN & PROFESSIONAL EXPERIENCE

UVA in Greece, Creative Interaction Design, *UI/UX Designer*, Heraklion, Greece **May - June 2025**

- Participated in a design-led research fellowship to investigate **multimodal interaction** for cultural heritage learning; developed "Olympet," a high-fidelity system comprising an AI-enabled interactive physical artifact and a synchronized mobile application
- Executed an **ethnographic field study** involving 20+ hours of contextual inquiry, stakeholder interviews, and **in-situ usability testing** with diverse user groups to evaluate engagement in non-traditional learning environments
- Applied **thematic analysis** to qualitative data to derive user personas and behavioral models, translating complex cultural requirements into a **cross-platform interaction framework**

The Spill Teem, *Data Research and Storyteller Intern,* Charlottesville, VA **January - May 2025**

- Analyzed multi-industry datasets (300+ companies, 1.4M+ employees) to extract patterns in organizational behavior, applying behavioral psychology and HCD frameworks to build structured research models
- Translated quantitative findings into executive-level narratives and structured documentation, guiding strategic decision-making and workshop development across multiple sectors

UVA Drama Department, *Assistant Projection Designer* **May 2026 - Present**

- Supporting the lead projection designer on the Drama Department's Fall 2026 mainstage production "*Are You Now or Have You Ever Been*"

TheCourseForum, *UI/UX Designer,* Charlottesville, VA **Sep 2025 - Dec 2025**

- Redesigned course browsing and detail pages to improve filter discoverability and integrate live instructor data, streamlining course selection for 10,000+ UVA students each semester

TEACHING & LEADERSHIP EXPERIENCE

UVA Computer Science Department, *Teaching Assistant* **August 2025 - May 2026**

- Mentored 10+ student teams across **two HCI courses**, guiding them to translate ambiguous problem spaces into research-grounded design decisions and functional prototypes
- Facilitated weekly sprint check-ins and calibrated rubric-based feedback, pushing student work toward methodological rigor across claims analysis, information architecture, and usability evaluation

LingXi Chinese Theater Club, *President, Director, Stage Manager & Graphic Designer* **Sep 2023 - present**

- Managing full-scale theatrical productions for a 30+ member organization, spearheading artistic direction, promotional graphic design, and operations from venue contracting through financial reconciliation

ACM CHI Conference on Human-Computer Interaction, *Student Volunteer,* Barcelona, Spain **April 2026**

- Supported conference operations at the premier international HCI venue, assisting sessions and attendee logistics across a 3,000+ researcher event